

DA5000W Linear algebra and optimization

Assignment 3

December 2025

Topics:

1. Least squares
2. Orthogonality
3. Gram Schmidt

Question 1:

The floor V and the wall W are not orthogonal subspaces, because they share a non-zero vector (along the line where they meet). No plane V and W in $\mathbb{R}^3$ can be orthogonal. Find a vector in the column spaces of both matrices such that this will be a vector Ax and also $B\hat{x}$. Think 3×4 matrix $[A \ B]$

Question 2:

Suppose there are eight vectors $r_1, r_2, n_1, n_2, c_1, c_2, l_1, l_2$ in $\mathbb{R}^4$

1. What are the conditions for those pairs to be bases for the four fundamental subspaces of a 4×4 matrix?
2. What is one possible matrix A ?

Question 3:

Project the vector b onto the line through a . Check the error vector e is perpendicular to a .

1. $b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

2. $b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ and $a = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}$

Question 4:

Find orthonormal vectors q_1, q_2, q_3 such that q_1, q_2 span the column space of

$$A = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -2 & 4 \end{bmatrix}$$

which of the four fundamental subspaces contains q_3 ?

Question 5:

Find orthonormal vectors A, B, C by Gram-Schmidt method from a, b, c

$$a = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} \quad c = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

prove that A, B, C are bases for the vector perpendicular to $d = (1, 1, 1, 1)$

Question 6: Let

$$\mathbf{a} = (1, 2, -1), \quad \mathbf{b} = (2, -1, 1), \quad \mathbf{c} = (3, 1, 0) \in \mathbb{R}^3.$$

(a) Check which pairs among $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are orthogonal.

Question 7:

Let

$$\mathbf{v}_1 = (1, 1, 0), \quad \mathbf{v}_2 = (1, 0, 1), \quad \mathbf{v}_3 = (0, 1, 1)$$

in $\mathbb{R}^3$. Use the Gram-Schmidt process to compute an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ for $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Question 8:

Application of least squares for the regression problem.

Using the provided laptop price dataset ([Link](#)), apply Least squares method to predict the laptop price using the laptop features.

1. Code least squares method.
2. Identify and mention the input features (X) and target variable (Y).
3. Print the equation of the fitted hyperplane
4. Print the Mean Squared Error (MSE) which is, $\frac{\sum_{i=1}^N (Y_i - \text{pred}(X_i))^2}{N}$
5. Interpret the fitted hyperplane and write about your observation of the relationship between the input variables and output

Code submission guidelines

1. Prepare your assignment exclusively in a Jupyter Notebook (.ipynb format).
2. Ensure all code cells are executed and their corresponding outputs are visible.
3. Name the notebook as following: $\{\text{Rollnumber}\}_{LA_MFDS_Assignment_3.ipynb}$